Since the pole and zero locations are known, the transfer function can be written in the form of Eq. 33-4 by simply plugging in the values:

$$H(z) = \frac{[z - (0.7071 + j\,0.7071)]\,[z - (0.7071 - j\,0.7071)]}{[z - (0.6364 + j\,0.6364)]\,[z - (0.6364 - j\,0.6364)]}$$

To find the recursion coefficients that implement this filter, the transfer function must be rearranged into the form of Eq. 33-3. To start, expand the expression by multiplying out the terms:

$$H(z) = \frac{z^2 - 0.7071z + j\,0.7071z - 0.7071z + 0.7071^2 - j\,0.7071^2 - j\,0.7071z + j\,0.7071^2 - j^2\,0.7071^2}{z^2 - 0.6364z + j\,0.6364z - 0.6364z + 0.6364^2 - j\,0.6364^2 - j\,0.6364z + j\,0.6364^2 - j^2\,0.6364^2}$$

Next, we collect like terms and reduce. As long as the upper half of the z-plane is a mirror image of the lower half (which is always the case if we are dealing with a *real* impulse response), all of the terms containing a "j" will cancel out of the expression:

$$H[z] = \frac{1.000 - 1.414z + 1.000z^2}{0.810 - 1.273z + 1.000z^2}$$

While this is in the form of one polynomial divided by another, it does not use negative exponents of z, as required by Eq. 33-3. This can be changed by dividing both the numerator and denominator by the highest power of z in the expression, in this case, z^2:

$$H[z] = \frac{1.000 - 1.414z^{-1} + 1.000z^{-2}}{1.000 - 1.273z^{-1} + 0.810z^{-2}}$$

Since the transfer function is now in the form of Eq. 33-3, the recursive coefficients can be directly extracted by inspection:

$$\begin{aligned} a_0 &= 1.000 \\ a_1 &= -1.414 \qquad b_1 = 1.273 \\ a_2 &= 1.000 \qquad b_2 = -0.810 \end{aligned}$$

This example provides the general strategy for obtaining the recursion coefficients from a pole-zero plot. In specific cases, it is possible to derive